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> Applicable environment: Jetson Orin NX, JetPack 7.2.1, DeepStream 9.1.

1. Case goal and processing flow

VehicleTypeNet classifies the cropped region of a vehicle; it does not replace the vehicle detector. The previous stage TrafficCamNet produces the vehicle boxes, DeepStream crops them and sends them into the vehicle model classification model.

```
Video/USB/CSI → TrafficCamNet → the vehicle crop → VehicleTypeNet
              → the model category is overlaid → the preview/the video output
```

2. The checks before running it

```
cd ~/deepstream_tao_apps/deepstream_app_tao_configs

# Confirm the configuration and the input source
grep -E "uri=|config-file=" run_vehicletype.txt

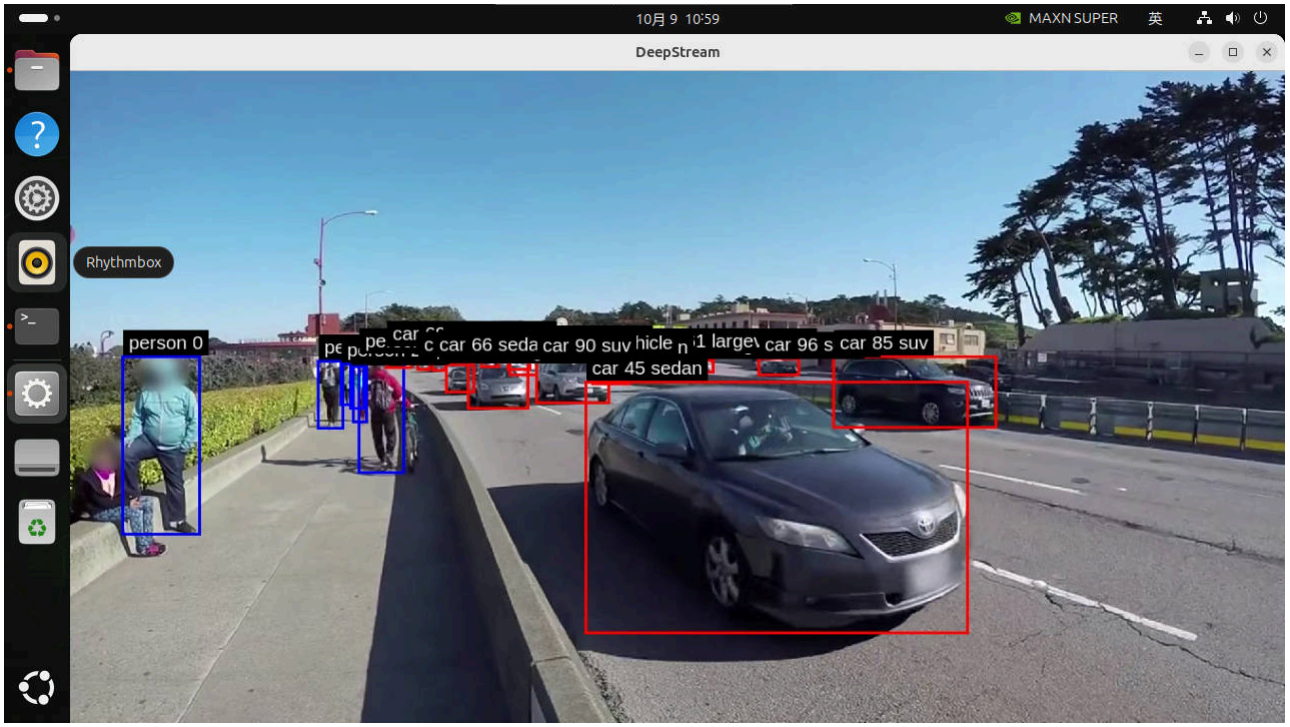
# Confirm that all the config-file paths are valid (the path trap was hit before)
for f in run_vehicletype.txt run_vehicletype_usb.txt run_vehicletype_csi.txt; do
  echo "--- $f ---"
  grep "config-file=" "$f" | while read line; do
    path=$(echo "$line" | cut -d= -f2)
    [ -f "$path" ] && echo "OK   $path" || echo "MISS $path"
  done
done
```

When you need the real-time camera, confirm that the camera is connected and that it is not used by another application.

3. Start the detection

To run the vehicle model classification alone (the video version):

```
export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh vehicletype video
```



The input is the sample_720p.h264 that comes with DeepStream. Running the cascade combination `run_vehicles.txt` runs the brand and the model classification at the same time.

When you replace the test video, modify the `uri=file://` of the `[source0]` section in `run_vehicletype.txt` to the absolute path of the MP4/H.264 file on the Jetson. Use `gst-discoverer-1.0` to confirm that the encoding can be decoded.

4. The real-time detection with a USB camera

Connect the USB camera, look at the device and the formats and then start the vehicle model branch:

```
ls -l /dev/video*
v4l2-ctl -d /dev/video0 --list-formats-ext

export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh vehicletype usb
```



The script detects the USB camera node automatically through `~/deepstream_tao_demo/find_usb_cam.sh`.

Watch the real-time vehicle boxes, the Tracker ID and the model labels in the Jetson desktop/VNC. The camera entry only previews and it does not produce an offline MP4.

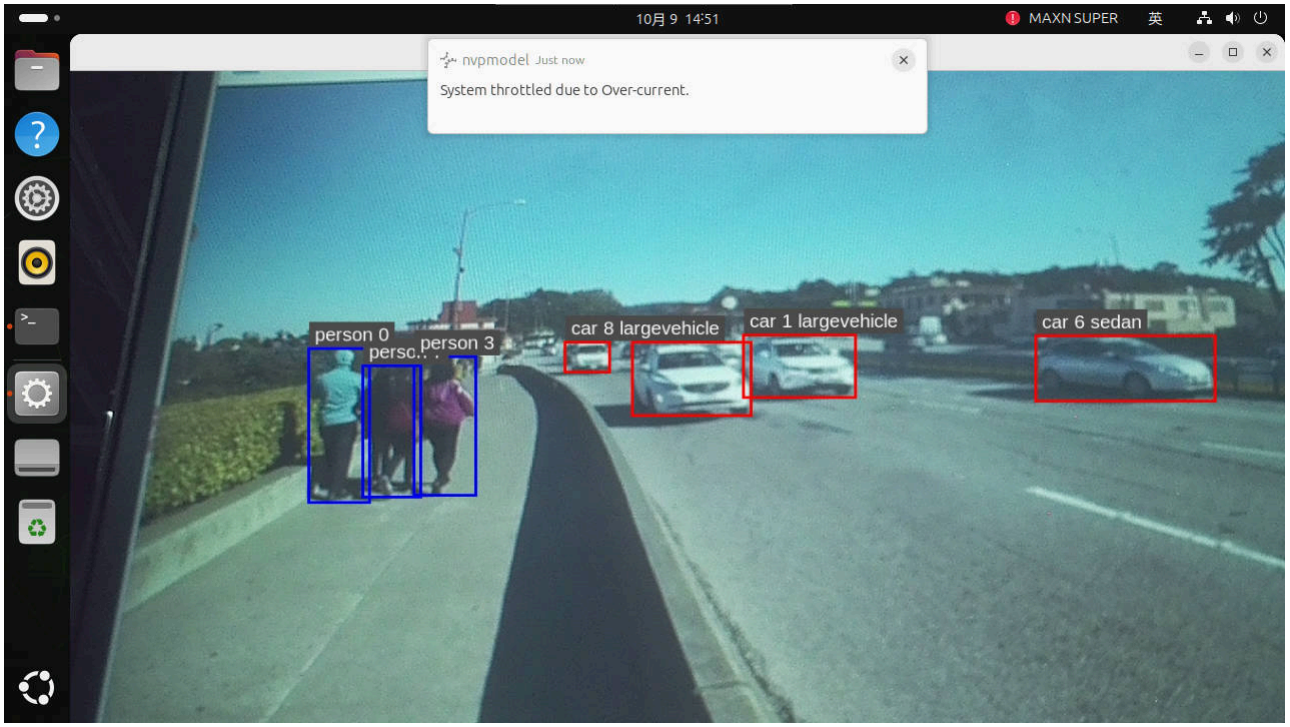
To stop it, press `Ctrl+C` once and wait for the application to exit and for the camera to be released. The model label is only produced after TrafficCamNet has detected a vehicle.

5. The real-time detection with a CSI camera

Connect a compatible camera, confirm that the driver and the Argus service work normally and run the following in a graphical terminal of the Jetson desktop/VNC:

Start:

```
export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh vehicletype csi
```



The CSI mode supports the real-time preview and it does not produce an offline video output. To stop it, press `Ctrl+C` once and wait for the program to release the camera.

6. Frequently asked questions

Symptom	The check and the handling
There is only `car + ID` and no model label	Check `network-type=1`, `is-classifier=1`, `operate-on-gie-id=1` and `operate-on-class-ids=0` in the vehicle model configuration, and confirm that the second-level engine is b4.
`Failed to load config file`	Start it with `run_case.sh`; all three entries have been unified to the `config-file=` path of TrafficCamNet + Tracker + VehicleTypeNet.
The model does not match the expectation	The set of the categories is limited and it depends on the region/the distribution of the data; evaluate it with the target material first.
The USB device cannot be opened	Check `/dev/video*`, the supported formats, the permissions and the occupancy.
There is no image with CSI	Check the compatible module, the ribbon cable, the sensor driver, the Argus service and whether the camera is in use; this entry directly uses the Argus `type=5`.
There is no video file in the camera mode	The camera entry only previews; the video output is only accepted in the offline branch in which the file sink is configured.
`nvv4l2h264enc has no property named 'gpu-id'`	It is a known non-blocking warning of DS 9.1 and it does not affect the inference or the display.

7. References

- The NVIDIA DeepStream TAO application examples: https://github.com/NVIDIA/DeepStream/tree/main/src/apps/tao_apps
- The DeepStream TAO integration documentation: https://docs.nvidia.com/metropolis/deepstream/dev-guide/text/DS_TAO_integration.html